Final Project Proposal

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Title and Abstract

Mapping and Predicting Commodity Flows Through the US Freight Rail Network

We have collected weekly data on 22 commodity classes received by nine freight rail companies since 2014. This data was publicly available on the Surface Transportation Board website. The goal is to start our analysis with a single company and then scale the cleaning to modeling pipeline to encompass the remaining companies. By analyzing this data we want to determine which companies are outperforming their competition in any given commodity category. This can take the form of both dominance as a whole, or specialization and will be especially useful when comparing companies of different sizes. We want to then extend this analysis to using Time Series to forecast future performance. This would give us a model that we can then push further using Deep Learning techniques such as XGBoost to improve our predictive abilities. By the end of this project we want to have a working forecast model with a usable pipeline attached for future additional data.

Introduction

For this project, we are primarily going to be using data collected from the Surface Transportation Board (STB). This data is originally in weekly xlsx spreadsheets with data including the types and amounts of commodities processed through each rail company's network, filled vs placed orders, average terminal dwell times, etc (Figure 1). For this project we are focusing on the weekly commodity origination and receiving numbers. At first we are planning on focusing on the company BNSF but expect that our final models will be highly scalable to the rest of the reporting companies on the STB website. We think that this data could give us a great weekly snapshot of the flow of goods within the US. On top of this, the semi-regionality of the different companies should give us a rough approximation of the regional aberrations within the commodity flows.

Methods

Our data is all publicly available and easy to gather as it is on the STB website under "Reports and Data". The difficulty comes in transforming it into a usable format. Each xlsx file contains five spreadsheets with data that we don't plan on focusing on for this project. We need to isolate the weekly commodity data for each week and then combine the results into a usable long formatted dataframe. We are using beautifulsoup to scrape the data off of the internet and then using R to transform and create our final usable data.

At this point we are focusing on data collected from the company BNSF in order to get a proof of concept. Luckily, the data for every company follows the same format and reports the same data in the same timeframes, so the data transformation and modeling should be highly scalable to the other major companies.

From here we want to use time series modeling to be able to forecast which commodities growth and concentration for a given railroad company. We will likely use an ARIMA model to get some explainable outcomes before moving into a deep learning model such as XGBoost to improve our predictive powers if time allows.

Expected Outcomes

If everything in this project goes accordingly we will have performed time series analysis on weekly commodity data from 2014-2026 for the nine major freight railroad companies that report to the STB. We will have gained insight into the transformation of the modern American supply chain over the last decade and have a model that will allow us to forecast the future shifts of the listed commodities. We will be able to determine which companies outperform in given commodity classes and extrapolate what this means for the regions that these companies are concentrated in.

By the end of this we will all have gained valuable experience in data gathering and wrangling, modeling using state-of-the-art techniques, and gained valuable insight into the realities of working as Data Scientists.

Group Considerations

Data Collection: Megan (40%), James (30%), Sakina (30%)

Data Wrangling: Sakina (60%), Megan (20%), James (20%)

Data Modeling: James (40%), Sakina (30%), Megan (30%)

Figure + Table Creation: Sakina (33%), Megan (33%), James (33%)

Poster Presentation: James (60%), Megan (20%), Sakina (20%)

Figures

11. i. Weekly Originated and Received Carloads by 22 Commodity Categories

Railroad	Commodity Group	Commodity Group Code	Originated	Received
	Chemicals	A	15,766	3,230
	Coal	B	13,853	8
	Coke	C	868	83
	Crushed Stone, Sand and Gravel	D	4,504	142
	Farm Products except Grain	E	149	102
	Food and Kindred Products	F	2,445	1,410
	Grain Mill Products	G	3,272	258
	Grain	H	6,117	1,139
	Iron and Steel Scrap	I	1,031	95
	Lumber and Wood Products	J	1,099	424
	Metallic Ores	K	168	94
	Metals	L	1,179	1,100
	Motor Vehicles and Equipment	M	1,804	4,522
	Non Metallic Minerals	N	356	182
	Petroleum Products	O	3,642	1,439
	Primary Forest Products	P	143	9
	Pulp, Paper and Allied Products	Q	679	741
	Stone, Clay and Glass Products	R	2,013	469
	Waste and Scrap Materials	S	474	240
	All Other	T	2,627	959
	Containers	U	56,971	8,463
	Trailers	V	1,492	10

em Nos. 1-6) Grain Loadings (Item No. 7) Grain Car Order (Item No. 8) Coal & Grain Plans (Items 9-10) Carloadings (Item 11)

Figure 1: A screenshot of the originally formatted xlsx file, showing an example of the weekly commodity data that will be used for this project as well as the additional spreadsheets that will need to be cleaned and removed.

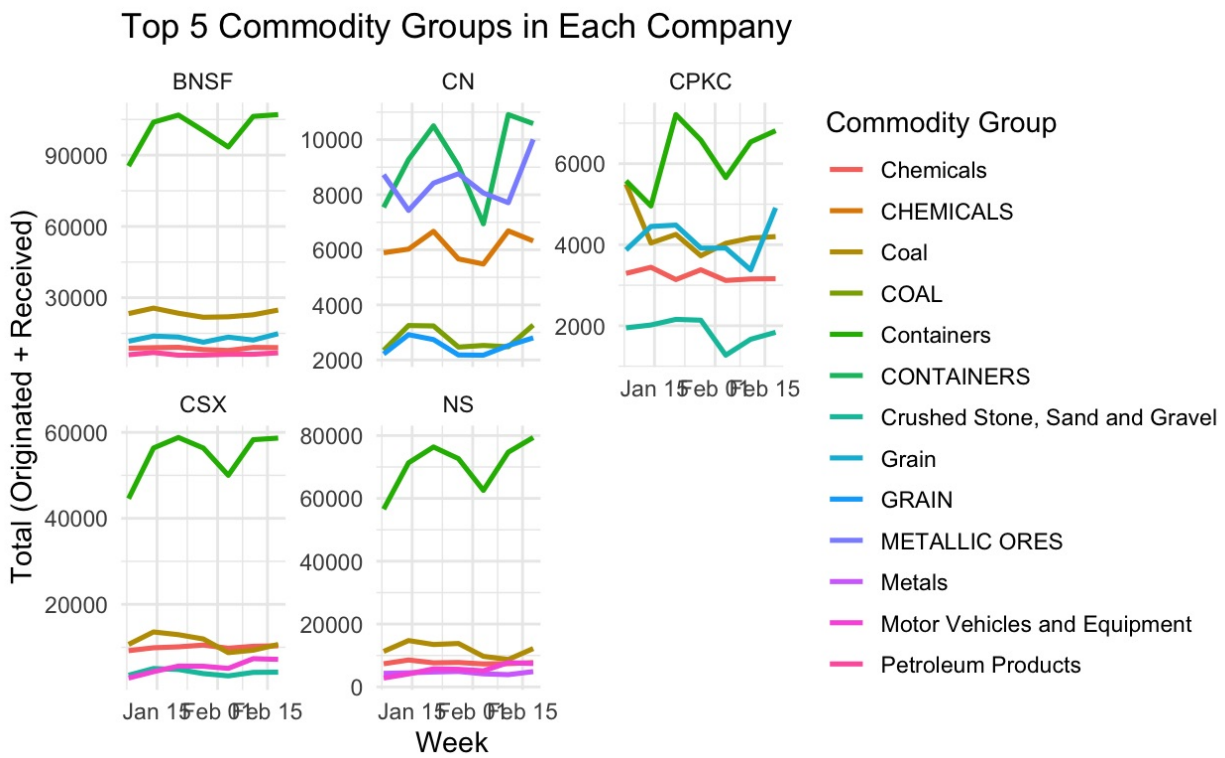

Figure 2: This shows the top commodity groups for five different railroad companies within the US and Canada. The x-axis is weekly, and includes data from the beginning of 2026 to the present. The y-axis is the commodity totals for each category.

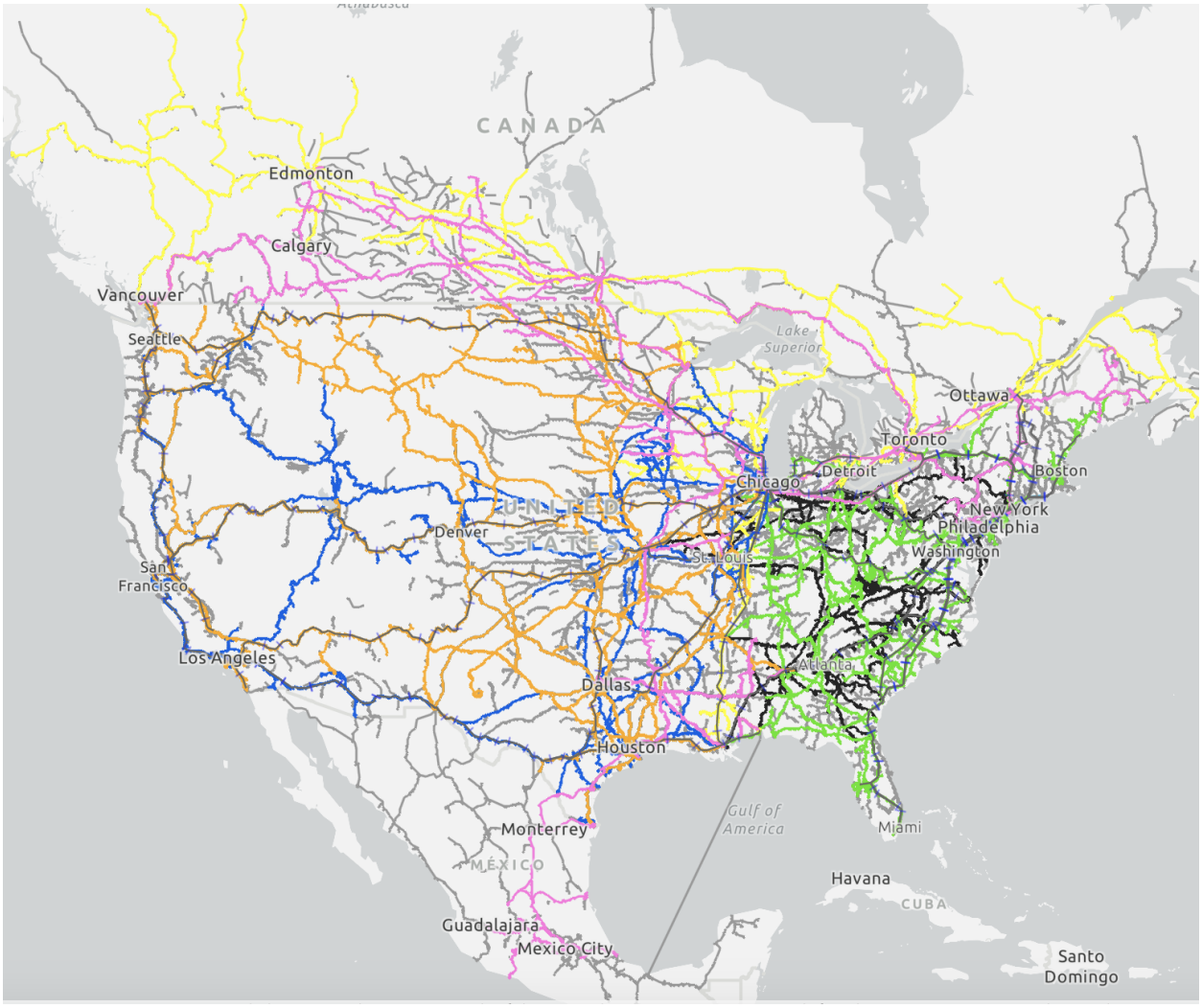

Figure 3: This is a map of North American railroads, color coordinated by railroad ownership.